MD. NAFIUL HAQUE NAFEE

Computer Science Undergraduate
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PROFESSIONAL PROFILE

Computer Science undergraduate at BRAC University with practical experience in artificial intelligence, machine learning, data analytics, natural language processing, computer vision, and digital application development. Experienced in developing and evaluating AI/ML solutions using Python, PyTorch and Scikit-learn with project work spanning text classification, computer vision, predictive analytics, Android applications, and full-stack web development. Interested in responsible AI, digital transformation, data-driven decision-making, innovation, and technology-enabled development.

EDUCATION

BRAC University — Dhaka, Bangladesh January 2023 – Present
Bachelor of Science in Computer Science | Current CGPA: 3.14/4.00
Expected graduation: January 2027

Matsuo-Iwasawa Laboratory, The University of Tokyo — GCI World 2026 September 2026 – Present
Currently enrolled in the Global Consumer Intelligence (GCI) data science program covering Python, NumPy, Pandas, Matplotlib, machine learning, model evaluation, feature engineering, SQL, time series analysis, competitions, and practical business applications.

BAF Shaheen College Kurmitola — Dhaka, Bangladesh July 2019 – December 2021
Higher Secondary Certificate (HSC), Science | GPA: 5.00/5.00

Cambrian School and College — Dhaka, Bangladesh January 2017 – February 2019
Secondary School Certificate (SSC), Science | GPA: 5.00/5.00

SELECTED AI, DATA & DIGITAL PROJECTS

Undergraduate Thesis — Dialect-Aware Bangla QA with RAG & Routed Experts

- Developed a modular Bangla question answering framework supporting Standard Bangla, Chittagonian, Sylheti and romanized input using dialect normalization, domain routing, RAG and routed LoRA experts on a shared 4-bit Gemma-4 model; closed 74% of the end-to-end dialect QA gap for both Chittagonian and Sylheti.

NLP • LLMs • RAG • QLoRA • Information Retrieval • Model Evaluation

Predictive Analytics & Explainable Machine Learning

- Developed a GPU accelerated XGBoost pipeline using approximately 95000 Steam titles to predict measurable recommendation tractions where the model achieved a 0.76 F1-score and 0.85 PR-AUC, incorporation validation-based threshold tuning, strict-versus-full feature evaluation and permutation importance analysis.

Data Analytics • Predictive Modelling • Explainable AI • SHAP

Multi-Task Computer Vision & Deep Learning

- Developed a multi-task U-Net–DenseNet121 model for pet segmentation and 37-breed classification; achieved 92.4% mIoU for segmentation and 89.3% test accuracy for classification.

Computer Vision • Deep Learning • PyTorch • Image Segmentation

News Classification Using NLP and Deep Learning

- Developed a four-class news classification pipeline comparing TF-IDF and Skip-gram Word2Vec representations with machine learning and recurrent neural network models; achieved 91.9% test accuracy and Macro-F1 using a GRU-based model.

NLP • Machine Learning • Deep Learning • Python • Model Evaluation

Full-Stack & Android Application Development

- Developed Android applications including Music Store and Fitness Care, alongside full-stack web applications using the MERN stack, gaining practical experience in digital product development and implementation.

Digital Products • Android • MERN Stack • JavaScript

NHT Fitness — Fitness, Nutrition & Motion-Tracking Android App

- Developed a native Kotlin/Jetpack Compose fitness application with workout planning and set tracking, nutrition and food search, hydration and body measurements, reminders, exercise-specific GIF guides, Room/Firebase persistence, and an experimental CameraX/ML Kit push-up repetition counter.

Android • Kotlin • Jetpack Compose • CameraX • ML Kit • Room • Firebase

Fire / Water / Lightning — Gesture-Controlled Android Game

- Developed a camera-based hand-gesture spell-dueling game using CameraX and MediaPipe, recognizing drawn circle, triangle, lightning-bolt and star gestures for elemental combat, with app flows for authentication, lobbies, matches, tournaments, friends, leaderboard and progression.

Android • Kotlin • Jetpack Compose • Computer Vision • MediaPipe • Firebase

Six Seven — Camera-Controlled Hand-Motion Game

- Developed a native Kotlin/Jetpack Compose Android game using CameraX and MediaPipe HandLandmarker to track two-hand movement and score alternating hand-height swaps, with smoothing, hysteresis/debounce, timed gameplay, feedback, leaderboard, audio and haptics.

Android • Kotlin • Jetpack Compose • CameraX • MediaPipe • Firebase

TECHNICAL SKILLS

Programming: Python, Java, Kotlin, JavaScript
AI & Machine Learning: Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Predictive Analytics, Explainable AI
ML & Data Libraries: PyTorch, Scikit-learn, Pandas, NumPy
Software Development: Android Development, MERN Stack, Full-Stack Development
Tools: Git, GitHub, Jupyter Notebook, Visual Studio Code

LEADERSHIP & EXTRACURRICULAR EXPERIENCE

ATOMIC BAFSK Science Club Publicity Manager | August 2019 – June 2021

- Managed publicity activities and supported communication and outreach for the science club.
- Contributed to organizing the 2019 Olympiad Competition and worked collaboratively with club members on event activities.

LANGUAGES & AREAS OF INTEREST

Languages: Bangla — Native | English — Proficient
Areas of Interest: Digital Transformation, Artificial Intelligence & Responsible AI, Data Analytics, Digital Innovation, Data Governance, Digital Product Development, Systems Transformation, Technology for Sustainable Development